COLLEGE OF MEDICINE

* The numbers like millions and thousand are not impo, percentages are kinda worthfull to know where disease is common not exact value

Vector-Borne infections

Prepared by
Assistant professor
Dr Shahow Abdulrehman Ezzaddin

Objectives

- To identify the common vectors and common vector borne infectious diseases
- To be able to diagnoses and manage common vector borne infections
- To be able to plan to control the known vector borne infections
- To plan for prevent many infectious diseases transmitted by vectors

Vectors

- Vectors are living organisms that can transmit infectious pathogens between humans, or from animals to humans.
- Many of these vectors are bloodsucking insects, which ingest disease-producing (Pathogen) microorganisms during a blood meal from an infected host (human or animal) and later transmit it into a new host, after the pathogen has replicated. → inside the vector but not necessarly it may just replicate in the host

COLLEGE OF MEDICINE

Vector-Borne Disease Transmission

→ Pathogen enter blood/ body of vector

- **Biological**: Most significant mode of transmission - Arthropod ingests a pathogen while taking a blood meal from an infected host - Pathogen multiplies within the arthropod (reservoir) –
- Pathogen is transmitted to another host when arthropod takes another blood meal

COLLEGE OF MEDICINE

Vector-Borne Disease Transmission

pathogen on outer surface of vector

pathogen does not multiply whilst being carried, chat GPT

- **Mechanical:** Vector physically carries pathogens from one place or host to another, usually on body parts or through the gastrointestinal tract {Housefly and cockroaches}

the vector feed on contaminated material like garbage, swallow pathogen and without pathogen being multiplied it will defecate or regurgitated on food or surfaces

Epidemiology

- Vector-borne diseases account for more than **17%** of all infectious diseases, causing more than **700, 000 deaths** annually.
- They can be caused by either **parasites**, **bacteria** or **viruses**. infective agent
- Malaria is a parasitic infection transmitted by female **Anopheles mosquitoes**.
- **Dengue** fever is the most prevalent widespread viral infection common in Africa transmitted by **Aedes mosquitoes**.

COLLEGE OF MEDICINE

Epidemiology

the common location of diseases are impo for travel Hx

Other viral diseases transmitted by vectors include:

- **Chikungunya fever**
- **Zika virus fever**
- **Yellow fever**
- **West Nile fever**
- **Japanese encephalitis** (all transmitted by mosquitoes)
- **Tick-borne encephalitis** (transmitted by ticks).

COLLEGE OF MEDICINE

Epidemiology

Many of vector-borne diseases are **preventable**, through:

-Protective measures → prevent bite by clothing, bed nets, insects repellent, vaccination

-Control of vector → remove vector

- Community mobilization. → awareness

Vectors

We can classify vector borne diseases into:

1- Mosquito borne

2- Tick borne

3- Fleas borne

4- Flies borne

5- Lice borne

COLLEGE OF MEDICINE

memo

Agents and vectors

Disease	Causative agent	Vector	Method of infection
Malaria	Plasmodium	Anopheles mosquito	Bite
Dengue fever	Virus	Aedes Mosquito	Bite
Encephalitis inflammation of brain	Virus	Culex mosquito	Bite
Chagas disease	Trypanosoma Cruzi	Triatomine bugs	Bite
Dysentery (amoebic)	Protozoa	Housefly	Contamination of food
Dysentery (Bacillary)	Bacteria (Shigella)	Housefly	Contamination of food
Cholera	Vibrio cholera	Housefly	Contamination of food
Leishmania	Leshmania protozoa (Tropica or Donovani)	Sandfly (phlebotomus)	Bite

memo

Agents and vectors

Disease	Causative agent	Vector	Method of infection
Plague	Bacterium (Yersinia pestis)	Rodent fleas	Bite by infected rodent
Rocky mountain fever	Ricketsia	Dog tick	Bite
Typhoid	Salmonella typhi (bacteria)	Houseflies	Contamination or Bite both bio and mechanical ? ask dr.shahow
Typhus	Ricketsia prowzeki	Fleas or lice	Contamination or Bite
Yellow fever	Yellow fever virus	Aedes mosquito	Bite
Chikungunya	Virus (Chikungunya virus)	Aedes mosquito	Bite
Schistosomiasis	Shistosomia (hematobium, mansoni or Japonicum)	Snail	Skin penetration

Yellow fever

- Yellow fever is an acute viral hemorrhagic disease transmitted by infected **mosquitoes**.
- The "yellow" in the name refers to the **jaundice** that affects some patients.
  caused hemolysis
- A small proportion of patients who contract the virus develop severe symptoms and approximately **half of those die within 7 to 10 days.**
- The virus is **endemic in tropical areas of Africa** and **Central and South America**.

Yellow fever:**Prevention**

1. **Vaccination:** The yellow fever vaccine is safe, affordable. Vaccine should not be given to infants aged less than 9 months;

- **A single dose** of yellow fever vaccine is sufficient to confer sustained immunity and **life-long protection** against yellow fever disease.
- A booster dose of the vaccine is not needed.
- The vaccine provides effective immunity **within 10 days** for 80-100% of people vaccinated, and within **30 days for more than 99% of people vaccinated.**

COLLEGE OF MEDICINE

Yellow fever:**Prevention**

city like area

2. Vector control: The risk of transmission in urban areas can be reduced by eliminating potential mosquito **breeding sites**, including by applying **larvicides** to water storage containers and other places where standing water collects. in other wordes Aedes mosquito live in still water

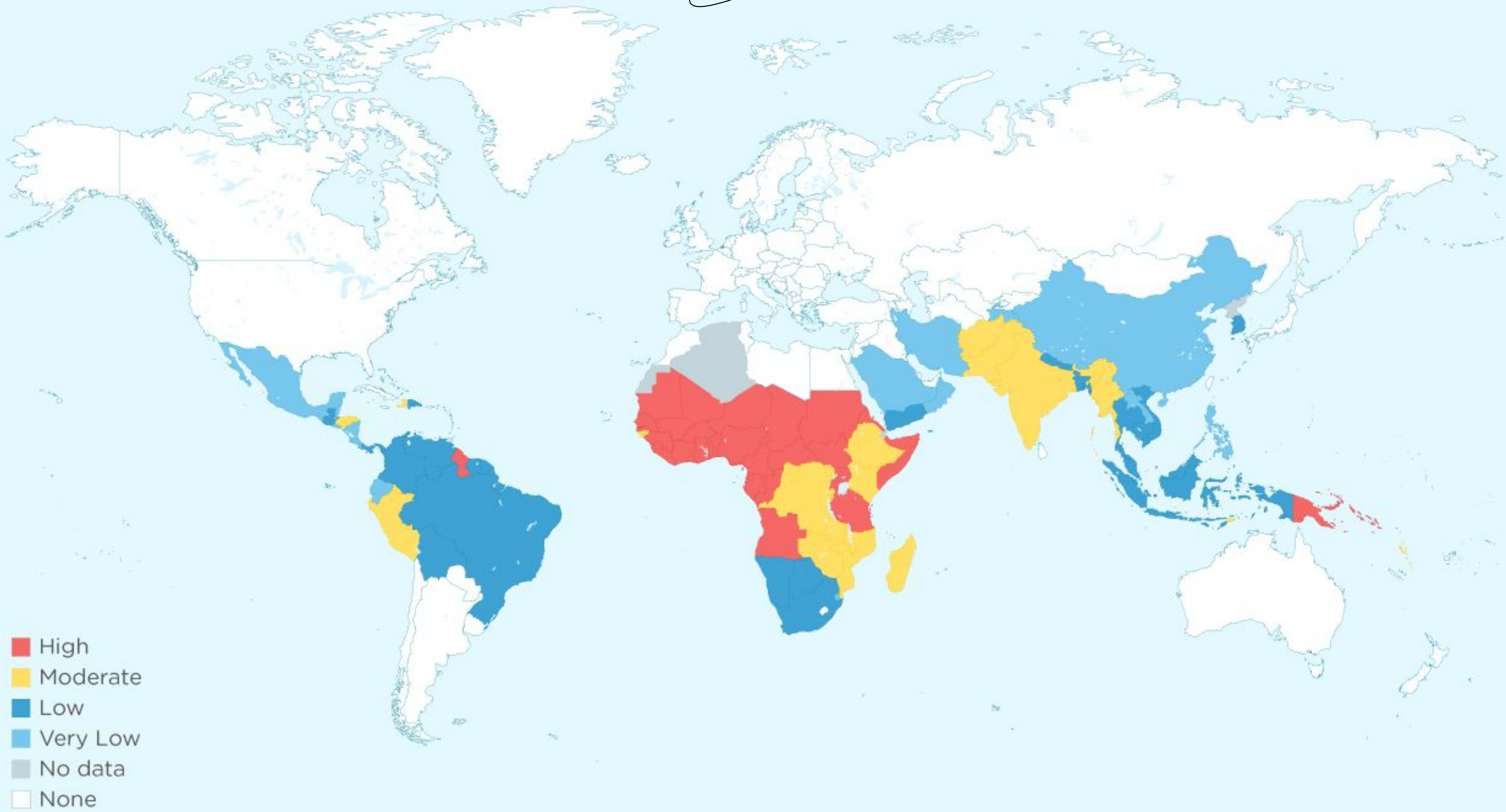
a type of insecticide used indoors and outdoors

Both vector surveillance and control are components of the prevention and control of vector-borne diseases.

Malaria (Epidemiology)

- Globally in 2022, there were an estimated 249 million malaria cases and 608 000 malaria deaths in 85 countries.
- The WHO African Region carries a high share of the global malaria burden as it was home to 94% of malaria cases (233 million) and 95% (580 000) of malaria deaths.
- Children under 5 accounted for about 80% of all malaria deaths in the Region

Malaria Heat Map
By CDC Estimated Risk



Data sourced from CDC, July 2018: https://www.cdc.gov/malaria/travelers/country_table/a.html

treated.com®

COLLEGE OF MEDICINE

Types of Malaria

plasmodium

- *P. vivax*
- *P. falciparum*
- *P. malariae*
- *P. ovale*

Diagnosis

- ***Microscopy :*** *Blood film for malaria (BFFM), both thin and Thick Film.*

  identify the specific type of malaria

  identify that we have malaria

- ***Serological test:*** The malarial fluorescent antibody test usually becomes positive two weeks or more after primary infection.

- ***Rapid diagnostic test (RDT).*** especially for people coming from prevelent countries

COLLEGE OF MEDICINE

used temporily not permanently since it has many side effects especially to eye

Chemoprophylaxis

so for travelers not residents due to S/E

it's the use of chemical agents usually drugs to prevent infection it can be use prior to or during or post-exposure but 100% prior to infection like in this case taking anti-malarial before having malaria

- Chemoprophylaxis is recommended for

  travelers from non-endemic areas and,

  as a short term measure for **soldiers, police** and

  labour forces serving in **highly endemic areas**.

Malaria vaccines

- **RTS**, S/A S01: it is recombinant (4 doses) not for Infant (6-12) ~~W.s~~ weeks , Efficacy (25-50) %
- The **R21 vaccine** is the second malaria vaccine recommended by WHO, following the RTS,S/AS01 vaccine, which received a WHO recommendation in 2021. (4th dose is given one year following the 3rd dose)

and all 4 must be taken to be effective

↳ so it's used for residents not travelers due to the time

Malaria vaccines

- **Both vaccines are shown to be safe and effective** in preventing malaria in children and, when implemented broadly, are expected to have high public health impact

Communicable diseases control program

1-Recognition of the infection

2-Notification

3-Finding the source of the infection

4-Finding the extent of the disease

Outbreak: Rapid sudden increase of number of cases in relevant to area and time

Communicable diseases control program

1-Recognition of the infection :- Diagnosis of the disease & confirmation of the diagnosis

Recognition depends on :

a-Clinical skill . **b-Lab diagnosis**

2-Notification :- Reporting the case to the local health authorities or the international health authorities to prevent the spread of infection * also Reporting outbreak

COLLEGE OF MEDICINE

Limitations of notification :-

A-Concealed (does not tell the truth) due to: Patient is afraid of isolation or Social stigma

B-Misdiagnosis of the correct disease due to over or under diagnosis
so to **overcome this problem by** Good skills and
especially in case of outbreak
Standard criteria for case definition

C-Incomplete reporting which can be overcome by
meaning the report form might be too boring and long to fill
simple forms

going backward to find cause

3-Finding the source of the infection: By

A-Good history of the travelling (patient movements)

B-Time sequence of event & consequences .

C-Incubation period of the disease .

Time between exposure of agent to onset of symptoms

4-Finding the extent of the disease.

how common/serious the disease in community

Management of any outbreak

it aimed to :-

- Reduction in the number of primary cases
- Reduction in the number of secondary cases
- Reducing the harm from the outbreak → morbidity and mortality
- Preventing a recurrence → by fixing the cause of disease

Managing an outbreak

1- Prepared an outbreak plan which have to reflects the situation , what is happening ,where & when did start

2- Call an urgent meeting of people who can help(Outbreak Control Meeting). The people with skills , responsibilities relevant to the situation .

Outbreak Control Meeting

It will result in:

1-Know the facts about the outbreak

2-Confirm an outbreak

3-Decide other information is required

4-Know signs & symptoms or laboratory results confirm cases → standard criteria

5-Decide on the appropriate forms of investigation & control

6-Decide who else needs to know

Q1) Acute poisoning of narcotics presents with:

A. Dilated pupils

B. Hypertension

C. Hyperventilation

D. Slow and shallow respiration

D

Q2) Regarding approach to chest pain:

a- almost all chest pain is ischemic in origin.

b- bronchitis is among the serious cause of chest pain.

c- Pain relief should be given within seconds to every patient with chest pain.

d- Typical angina pain is that pain which relieved by nitroglycerin.

D

Q3) regarding Measles:

a- the rash of measles is Maculopapula rash.

b- measles vaccine is recommended from 2 month of age.

c- measles virus is not highly contagious.

d- incubation period between few hours to 5 days.

A

Q4) Regarding Diphtheria

a) It is a chronic bacterial infection

b) The toxin of the virus causes myocarditis

c) The disease is of colder months

d) The cow is the reservoir

C

Q5) A 65-year-old man develops the onset of severe knee pain over 24 hours. The knee is red, swollen, and tender. The patient does not have fever or systemic symptoms. He has a history of diabetes mellitus and cardiomyopathy. Definitive diagnosis is best made by which of the following?

A. Serum uric acid

B. Serum calcium

C. Arthrocentesis and identification of positively birefringent rhomboid crystals

D. Rheumatoid factor

C

Q6) **What is the most likely diagnosis?**

a) Hemochromatosis

b) Rheumatoid arthritis

c) Systemic lupus erythematosus

d) Psoriatic arthritis

D

Q7) **Breast feeding is contraindicated in :**

a) Multiple birth

b) Neonatal Jaundice not due to breast milk allergy

c) Premature

d) Viral infection as HBV

D

Q8) Anti-motility not to be used for diarrhea in children under 5 years old because they:

a) Increase viral diarrhea.

b) Cause paralytic ileus.

c) Lead to oral thrush

d) May cause CNS symptoms

B

Q9) Latent period

a) The point at which the infectious agent enters the susceptible individual

b) The period from the time of infection until the infected individual is able to transmit the infection .

c) The period during which the infected person is able to transmit the infectious agent to other individuals

d) When the infectious period ends.

B

Q10) Which of the following can be a hint to Shigella infection :

a) Abdominal pain and vomiting.

b) Watery diarrhea, which proceed to bloody diarrhea with mucus

c) Fever and abdominal pain

d) Hypovolemic shock due to sever diarrhea

B